

Introduction to Probability

Describe how a sample space is used to estimate probabilities.

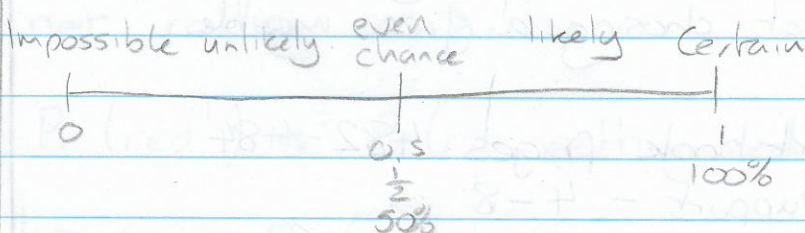
Questions

Notes

How can I write a probability?

The probability of an event is between 0 and 1. Can be written as a fraction, decimal or percentage.

What's a probability scale?



How do I estimate the probability?

If all outcomes are equally likely then

$$\Pr(\text{Event}) = \frac{\text{number of outcomes where the event occurs}}{\text{Total number of outcomes}}$$

What is a sample space?

A sample space is a list of all possible outcomes.

Examples →

e.g The sample space for rolling a 6-sided die is 1, 2, 3, 4, 5, 6

$$\Pr(\text{rolling a 5}) = \frac{1}{6}$$

↑ Probability ↑ event 6 - six possible outcomes.

$$\Pr(\text{rolling } > 4) = \frac{2}{6} = \frac{1}{3}$$

← two outcomes > 4

↑ 6 outcomes in total.

Summary

A sample space lists all the possible outcomes you can use this to see how many outcomes there are. To find the probability of an event you see how many outcomes for the event and divide by the total number of outcomes. The answer can be left as a fraction or changed to a decimal or percentage.